

Cost-Effectiveness Analysis

METIMMOX-1 Economic Evaluation

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1 Overview

This cost-effectiveness analysis evaluates standard of care against two pre-treatment biomarker-guided immunotherapy strategies for metastatic MSS/pMMR colorectal cancer:

- CRP-guided treatment selection
- TMB/BRAF-guided treatment selection

The economic survival model is a single joint model with CRP and TMB/BRAF treatment interactions. Economic strategies are restricted to pre-treatment biomarkers.

2 Economic Survival Model

Table 1: Single economic survival model

Component	Formula
OS	Surv(OSwk, Death) ~ Age + sex + Rx + crp*Rx + tmb_braf*Rx
PFS	Surv(PFSwk, Progression) ~ Age + sex + Rx + crp*Rx + tmb_braf*Rx
Control	Surv(time, event) ~ Age + sex

Table 2: Economic strategies

Strategy	Definition
Standard of Care	All patients receive standard FLOX chemotherapy
CRP-guided	CRP-positive patients receive FLOX + nivolumab; CRP-negative patients receive FLOX
TMB/BRAF-guided	TMB/BRAF-positive patients receive FLOX + nivolumab; TMB/BRAF-negative patients receive FLOX

3 Base Case Results

Table 3: Base case results at WTP = EUR 51,000

Strategy	Cost	QALYs	NMB
Standard of Care	EUR 20,867	1.414	EUR 51,254
CRP-guided	EUR 55,026	1.384	EUR 15,570
TMB/BRAF-guided	EUR 60,798	1.353	EUR 8,207

Table 4: Incremental cost-effectiveness results

Strategy	Cost	QALYs	Incremental Cost	Incremental QALYs	ICER	Status
Standard of Care	EUR 20,867	1.414	–	–	–	ND
CRP-guided	EUR 55,026	1.384	–	–	Dominated	D
TMB/BRAF-guided	EUR 60,798	1.353	–	–	Dominated	D

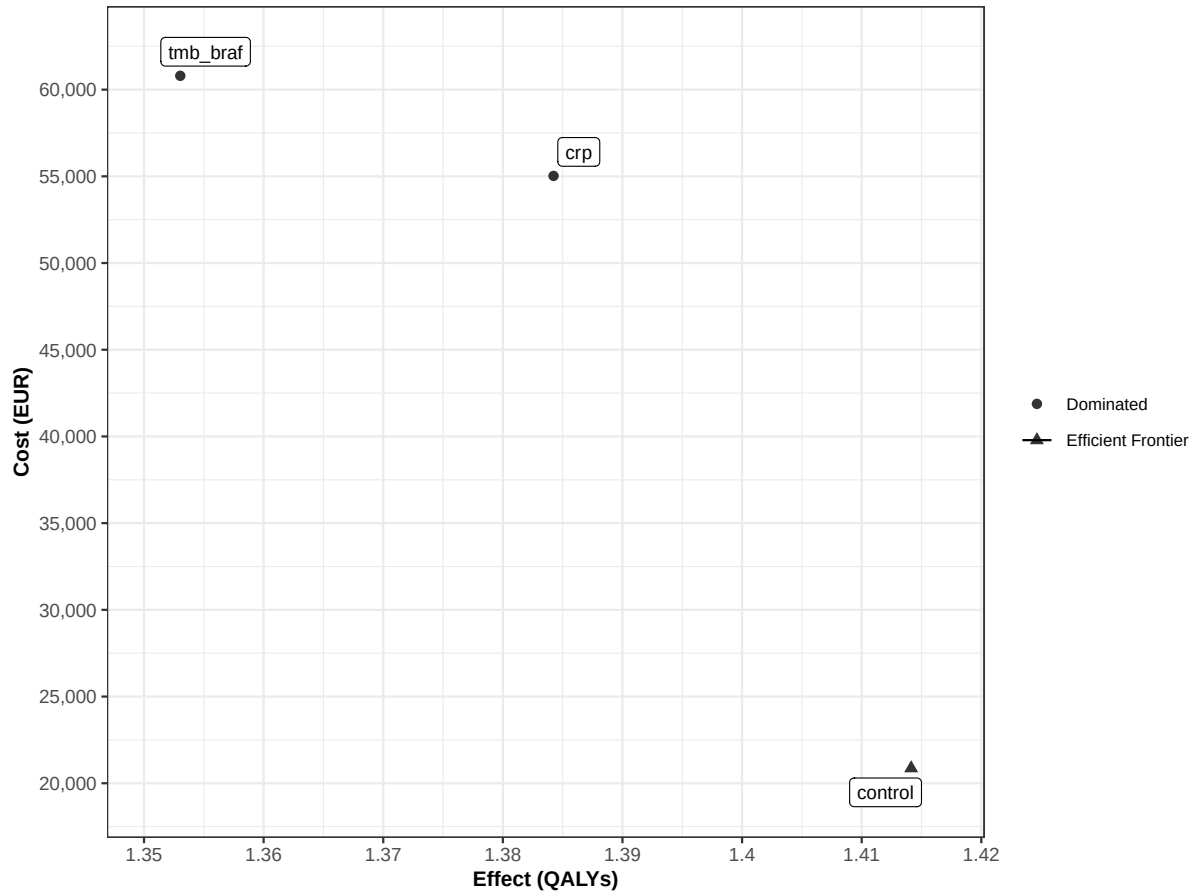


Figure 1: Cost-effectiveness plane for the single joint economic model

4 Probabilistic Sensitivity Analysis

Table 5: PSA summary at WTP = EUR 51,000

Strategy	Mean Cost	Mean QALYs	Probability Cost-Effective
Standard of Care	EUR 21,159	1.302	95.6%
CRP-guided	EUR 55,008	1.407	2.1%
TMB/BRAF-guided	EUR 60,705	1.389	2.2%

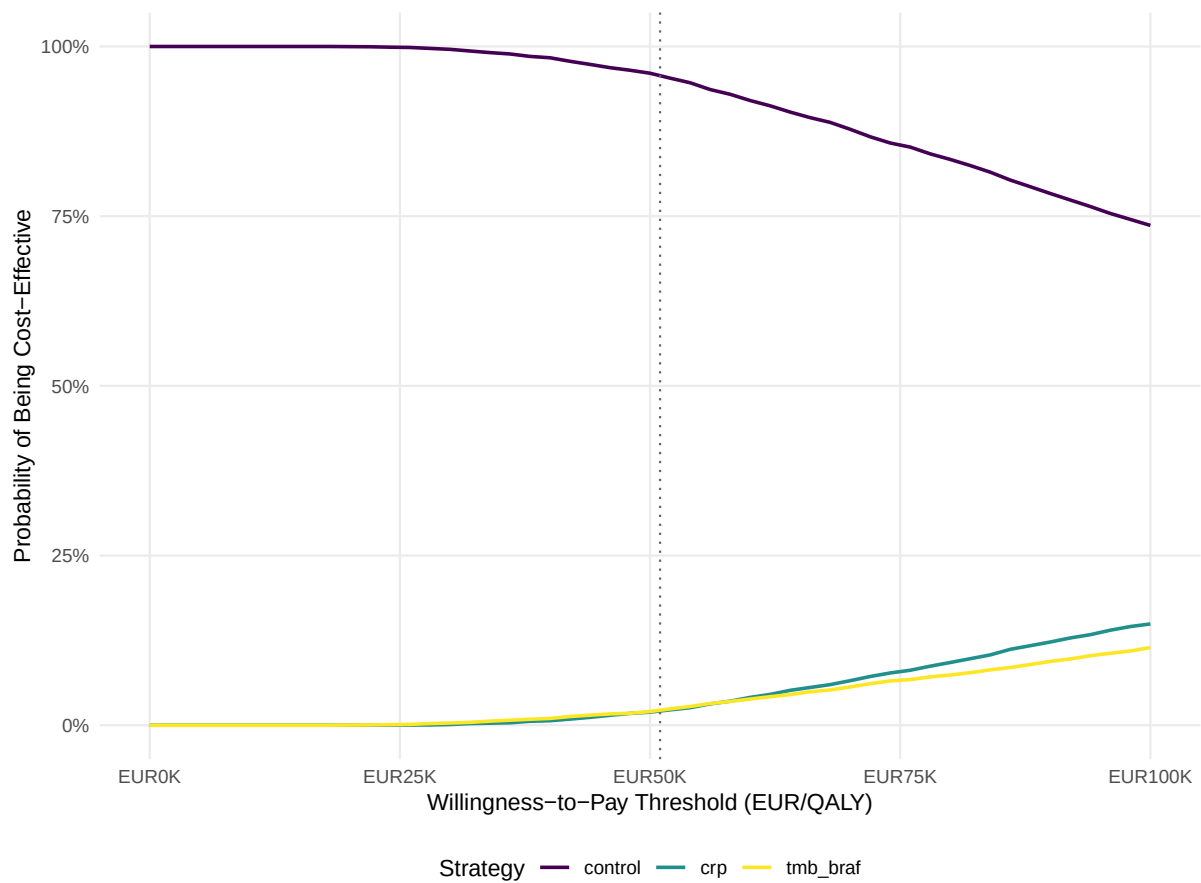


Figure 2: Cost-effectiveness acceptability curves

5 Summary

The economic evaluation now uses one joint survival model and three economic strategies: standard of care, CRP-guided immunotherapy, and TMB/BRAF-guided immunotherapy.

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